

Note: 1) All questions are compulsory.

2) Figures to the right indicate maximum marks.

3) Use of Non-programmable calculators is permitted.

4) Symbols used have their usual meaning.

Q.1) A) Attempt any one question.

[10 M]

- Describe how fixed bias is achieved for CE amplifiers. State its advantages and disadvantages.
- What is faithful amplification? State and explain the condition to be fulfilled to achieve faithful amplification in a transistorized amplifier.

Q.1) B) Attempt any two question.

[10 M]

- It is desired to set the operating point at 4V, 1.2mA by biasing the silicon transistor with collector feedback resistor R_B . If $\beta=120$, find the value of R_B .
- Explain the following terms a) Current gain (b) Voltage gain
- The input signal voltage of an amplifier is 2V and the input ascent is 1ms. The output current is 20 mA through a load resistor of 6000Ω . Determine input resistance of the amplifier.
- In collector feedback biased silicon transistor circuit, $R_C=1K\Omega$, $R_B=120K\Omega$ & $\beta=100$. Determine the operating point if $V_{CC}=+10V$.

Q.2) A) Attempt any one question.

[10 M]

- Explain the concept of differentiator with a neat labeled circuit diagram and obtain the expression of output voltage for the same.
- Draw neat labeled diagrams for the summing amplifier and obtain an expression for output voltage and state applications of the summing amplifier.

Q.2) B) Attempt any two questions.

[10 M]

- To have a gain of 25 for an inverting Op-Amp, what should be the feedback resistor required? Given input resistance is 400Ω .
- Write a short note on the critical frequency of the integrator.
- Explain the concept of frequency response of Op-Amp.
- What will be the maximum frequency that can be applied without distortion if the gain - bandwidth product of Op-Amp is 1 MHz Given: For negative feedback gain is 100.

Q.3) A) Attempt any one question.

[10 M]

- What is the shortcoming of R-S flip-flop? Explain how it is removed in JK flip-flops. Describe how the J-K flip-flop is realized. Describe its operation.
- Draw a block diagram of an up-down ripple counter. Explain its operation. Give its truth table for the countdown mode.

[P.T.O.]

Q.3) B) Attempt any two question.**[10M]**

- Draw the logic circuit of an R-S flip flop using NOR gates and write down its state transition table.
- Draw neat labeled diagram for master J-K flip flop and give its truth table.
- Explain the concept of Serial in serial out shift register (SISO).
- Give classification of different types of counters.

Q.4) Attempt any three questions.**[15M]**

- What do you understand by stabilization of the operating point? What is the definition of stability factor 'S'?
- Determine the open circuit voltage of the source, v_s to provide an amplifier input voltage v_i of 0.25 V when the internal resistance of the source is 50 ohms.
- In an inverting op-amp as an adder find the output voltage assuming the Op-Amp is initially nulled. If $R_1=4K\Omega$, $R_2=3K\Omega$, $R_3=8K\Omega$, & $R_f=10K\Omega$ for three input voltages $V_1=0.5V$, $V_2=-5.5V$ & $V_3=5V$ respectively.
- Write a note on the comparator.
- Differentiate between latch & flip flop
- What is the counter? What is meant by a serial counter and parallel counter?

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